OLIST · DELIVERY PERFORMANCE REVIEW

Late deliveries: what they cost, where they come from, and what to change

6.79% of delivered orders arrived after the promised date between Jan-2017 and Aug-2018, and 18.96% in the worst month. This report sizes the problem, finds where it comes from, and sizes three fixes.

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Executive summary: late orders come mostly from the carrier, in four states and three peak months

QUESTION	WHAT WE FOUND	WHAT IT MEANS
How big is it, and when does it spike?	6.79% of orders late. Nov-2017, Feb-2018 and Mar-2018 hold 48% of all late orders.	Not a promise-length issue: target months and places.
What does a late delivery cost?	1-2 star share 62.42% late vs 9.25% on time; repeat-buying drop not proven (p = 0.06).	The cost is reputation: track the 1-2 star share.
Where does it come from?	72.3% of late orders left the seller on time. 4 states hold 32.4% of 2018 late orders.	Logistics owns most of it, in four states.
What should Operations change?	Chosen on 2017, tested on 2018: up to **710**, **254** and **115** late orders avoided.	**Carrier target · +3 promise days · seller watchlist**

Business problem: about 1 in 15 orders arrives late, and two teams disagree on why

Olist, a Brazilian e-commerce company, promises a delivery date at checkout. Between Jan-2017 and Aug-2018, **6,531 of 96,203** delivered orders arrived after that date.

TWO VIEWS INSIDE THE COMPANY

Customer Experience

Late orders bring 1-star reviews.

Logistics

Sellers are slow to hand parcels to the carrier.

THE HEAD OF OPERATIONS ASKED

Q1 How big is the late-delivery problem, and when does it spike?

Q2 What does a late delivery cost us?

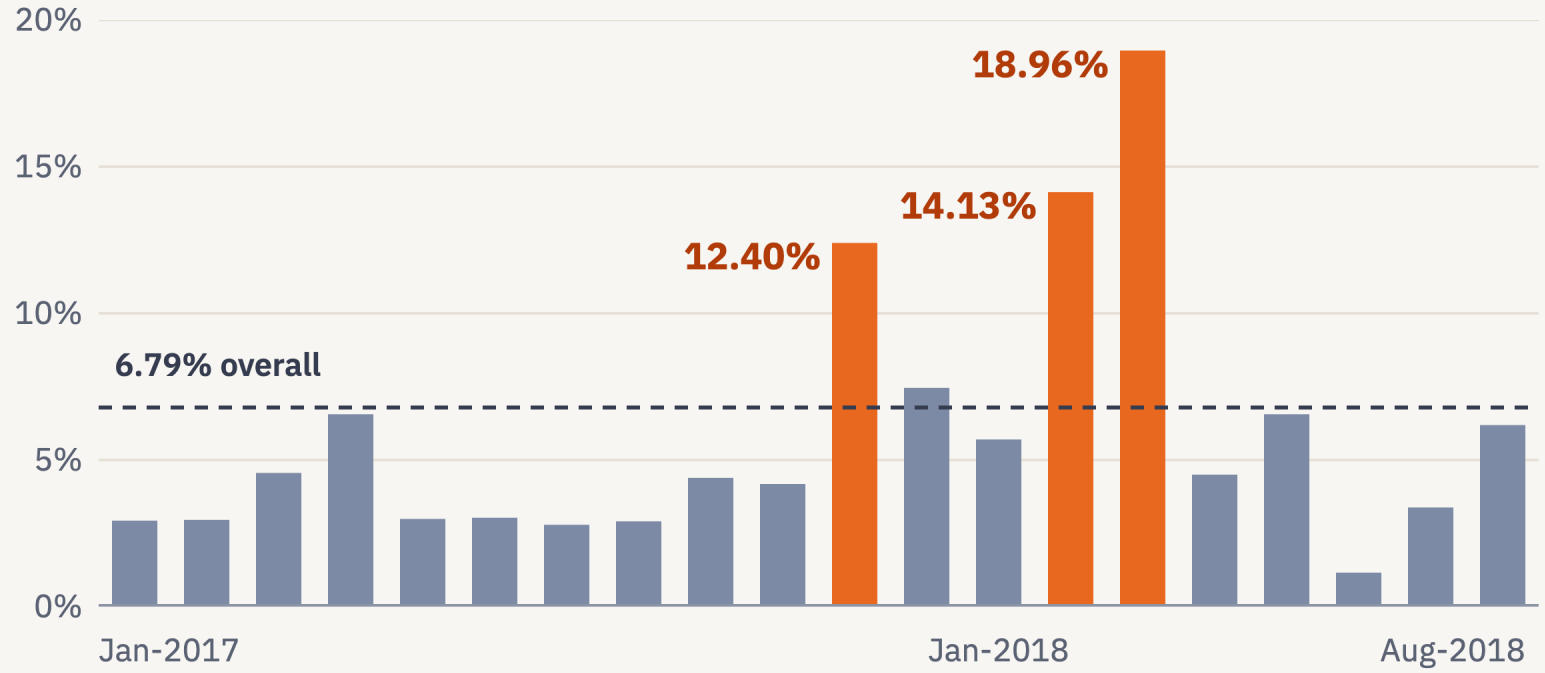
Q3 Where does it come from: the seller or the carrier, which states and routes?

Q4 What should Operations change, and what is each change worth?

Late = delivered on a calendar day after the promised date

Finding 1 · Lateness spikes in three months; in the other 17 it stays between 1.2% and 7.5%

Late rate by purchase month, Jan-2017 to Aug-2018 (orange: the spike months)



6.79%

late: 6,531 of 96,203 delivered orders

48%

of all late orders came in the 3 spike months

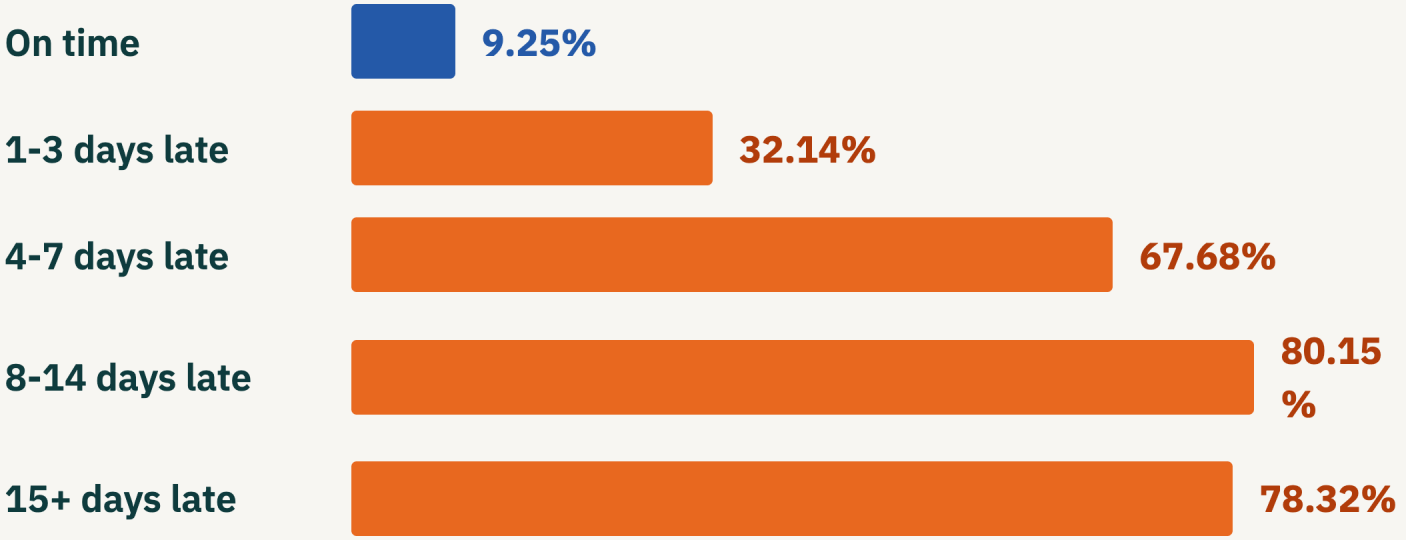
24 vs 10

median days promised vs days to deliver

Source: notebook 02, report_monthly

Finding 2 · A late order turns into a bad review, but a drop in repeat buying is not proven

Share of 1-2 star reviews by days late, with 95% confidence intervals



Average review: **4.29** stars on time vs **2.27** late. Every interval is within ±2.3 points, so the bands are clearly apart.

REPEAT BUYING IN 180 DAYS

1.96% after on time

1.57% after late

−0.38 points, 95% CI −0.71 to +0.02, p = 0.06: **not proven**.

Worst case, about **R$ 3,491 a year** in lost repeat orders.

Source: notebook 03, report_late_bands, report_repeat

Finding 3 · Most of the delay happens with the carrier, not the seller

Average days from purchase to delivery, split into the seller's part and the carrier's part



In the spike months the carrier slowed down; the sellers did not

Purchase month	Days to hand over	Days with carrier
May-Oct 2017 average	3.1	8.5
Nov-2017	4.0	11.1
Feb-2018	3.5	13.4
Mar-2018	3.3	13.0

72.3%

of late orders were handed to the carrier on time

21.13%

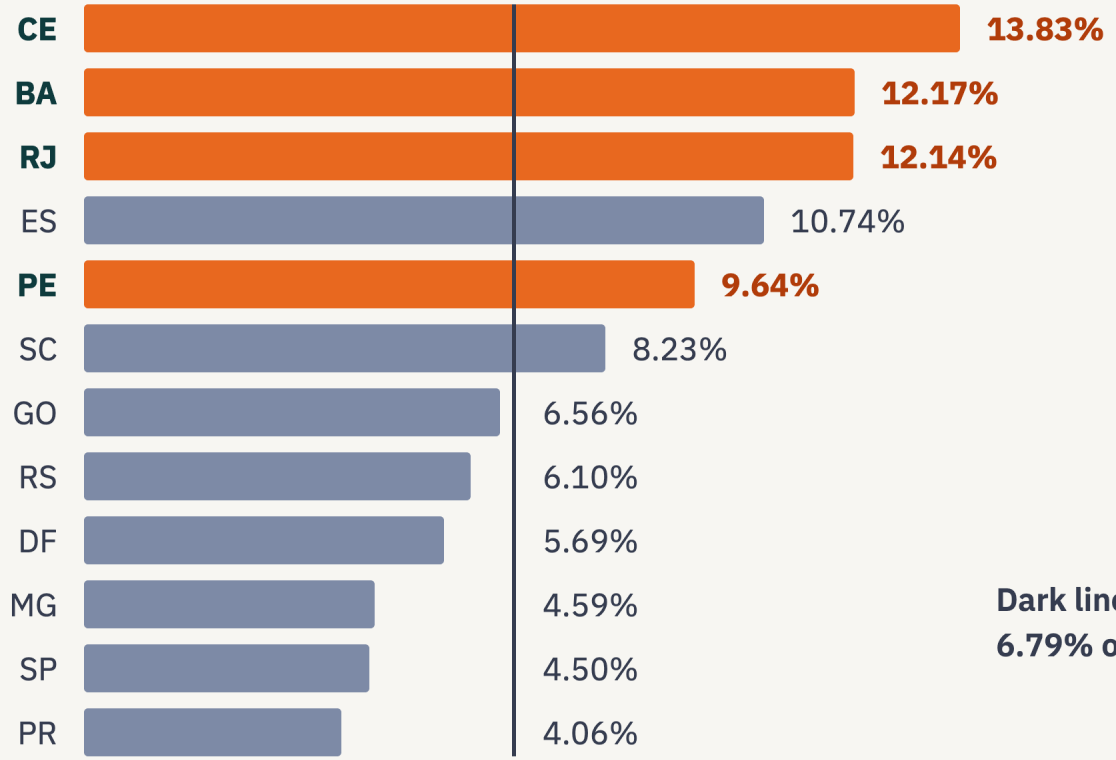
late when the seller hands over late, vs 5.39% when on time

Logistics' view holds for some orders, but sellers explain the smaller part.

Source: notebook 04, report_time_split, report_handover, report_spikes

Finding 4 · The same four states are late every year: BA, CE, PE and RJ

Late rate by customer state, states with 1,000+ orders, Jan-2017 to Aug-2018



32.4%

of 2018 late orders came from the four states, which had 18.1% of orders

8.1% vs 4.5%

late, orders to another state vs within the seller's state

1,152

late orders on one route, SP to RJ (14.15% late)

Orange: the four problem states. Source: notebook 04, report_states, report_routes

Insights · What the four findings mean for Operations

INSIGHT 1

The problem is concentrated, so the fix can be targeted

3 months hold 48% of late orders. 4 states hold 32.4% of 2018 late orders with 18.1% of orders.

So: act in those states and before the peak months, not with a longer promise for everyone.

INSIGHT 2

The carrier leg is the main lever, sellers the second

72.3% of late orders left the seller on time. In the spike months only the carrier days rose.

So: Logistics leads the fix. Seller Ops works on the smaller share that starts late.

INSIGHT 3

Lateness costs reviews, so reviews are the measure

62.42% of late orders get 1-2 stars. Repeat buying is rare (1.9%) and its drop is not proven.

So: judge each fix on the late rate and the 1-2 star share, not on repeat revenue.

Recommendations · Three fixes, chosen on 2017 orders and checked on 2018 orders they never saw

Late orders avoided out of 4,078 late orders in Jan-Aug 2018, and bad reviews avoided

1 · LOGISTICS

Carrier target for BA, CE, PE and RJ



up to 710 · 17.4%

378 bad reviews

2 · OPERATIONS

+3 promise days in the four states only



254 · 6.2%

135 bad reviews

3 · SELLER OPS

Watchlist sellers hand over on time



115 · 2.8%

61 bad reviews

The three overlap in the four states, so don’t add them up. Action 1 is a target, not a tested change: the data has no carrier name.

Recommendation 2 · Three extra promise days in the four states cut their late rate from 13.82% to 11.16%

Every Jan-Aug 2018 order in BA, CE, PE and RJ re-checked with a longer promise

Extra days	Late rate	Promise, days	Late avoided	Bad reviews avoided
0 (today)	13.82%	28.8	0	0
+2	11.93%	30.8	180	96
+3	11.16%	31.8	254	135
+5	9.74%	33.8	390	207
+7	8.05%	35.8	551	293

Late rate and promise are for the four states. Only they change: the national average promise moves from 23.4 to 24.0 days.

WHY +3 AND NOT +7

Every extra day is a longer wait shown at checkout, and some buyers may leave.

The data has no checkout data to measure that, so start in the middle and **test it on a share of orders**.

Source: notebook 05, report_promise_buffer

Action plan · One owner, a first step and a measure for each fix

1 · LOGISTICS

Carrier target for the four states

Impact

Up to 710 late orders, 378 bad reviews

First step

Share route late rates with the carrier, SP to RJ first

Measure

Late rate and carrier days in the four states

2 · OPERATIONS

+3 promise days in the four states

Impact

254 late orders, 135 bad reviews

First step

Test on a share of orders for two months

Measure

Late rate, 1-2 star share, checkout conversion

3 · SELLER OPS

Handover help for watchlist sellers

Impact

115 late orders, 61 bad reviews

First step

Send the 89 current sellers their own handover record

Measure

Late-handover rate of each watchlist seller

Next steps · The first 90 days

WEEKS 1-2 · AGREE

- Hand over the Excel files: watchlist, promise options, impact table
- Agree an owner and a target for each action
- Add the four-state late rate to the monthly review

MONTH 1 · START

- Logistics meets the carrier with the route data
- Start the +3-day test on a share of orders
- Seller Ops contacts the watchlist sellers

MONTHS 2-3 · DECIDE

- Compare test and control: late rate, 1-2 star share, conversion
- Decide whether to roll out +3 days
- Agree carrier capacity before the November peak

DATA TO ASK FOR

The **carrier of each order**, to test action 1, and **checkout data**, to measure what a longer promise costs.

Limitations · What we don’t know yet, and how we would check it

Limitation	How to check it
The data has no carrier name, so action 1 is a target, not a tested change	Get the carrier of each order; compare carriers on the same routes
A longer promise may lose buyers, and there is no checkout data	Test +3 days on a share of orders and compare conversion
Bad reviews avoided assume an avoided late order gets the on-time review rate	Track the 1-2 star share in the test group
Repeat buying is rare (1.9%), so a small real effect could be missed	Re-check with more months of data; the interval allows up to −0.71 points
The route uses the seller of the first item	Small: 1.3% of orders have 2+ sellers

CONCLUSION

Late deliveries cost Olist its reviews. Most of the delay is with the carrier, in four states and a few peak months, so that is where to fix it first.

Logistics

A carrier target for BA, CE, PE and RJ

Operations

Test +3 promise days in those states

Seller Ops

Work with the watchlist sellers

Track the late rate and the 1-2 star share every month on the Tableau dashboard.

Appendix · Data, method and definitions

Data

Table	Rows
orders	99,441
order_items	112,650
order_reviews	99,224
customers	99,441
sellers	3,095
products	32,951

Olist public dataset, Kaggle, CC BY-NC-SA 4.0. Analysis: 96,203 delivered orders, Jan-2017 to Aug-2018.

Method

1. PostgreSQL: load, 6 cleaning fixes, data-quality checks
2. Views and report tables, Wilson 95% CIs in SQL
3. Python notebooks: pandas, statsmodels, matplotlib
4. Two-proportion test and Newcombe interval for repeat buying
5. Out-of-time test: chosen on 2017, measured on Jan-Aug 2018
6. Tableau Public dashboard and Excel hand-off

Definitions

Late: delivered on a calendar day after the promised date

Bad review: 1 or 2 stars, latest review per order

Late handover: parcel reached the carrier after the seller's shipping limit date

Came back: a new order 1 to 180 days after the first

Problem state: 500+ orders in 2017, late on more than 8%